

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2021-P1-Model)** **Date: 04-02-2026**

Time: 09.00Am to 12.00Pm **GTA-14** **Max. Marks: 180**

04-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2021-P1)_GTA-14_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2021-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 10)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 11 – 16)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 23)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 24 – 29)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 30 – 35)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 42)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 43 – 48)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 49 – 54)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

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PHYSICS **Max Marks: 60**

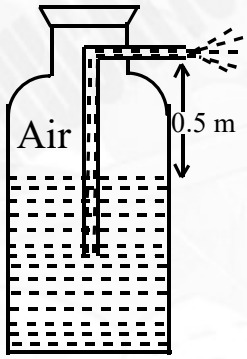
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases)

1. Water is forced out of a fire extinguisher by air pressure as shown in figure. Find the gauge pressure of air required in the tank so that water jet comes out with a speed of 30 m/s when the water level in the tank below the nozzle is maintained constant as 0.5 m is $\left[g = 10 m/s^2, \rho_{water} = 10^3 kg/m^3\right]$



A) $4.55\times10^5 N/m^2$ **B)** $5.55\times10^5 N/m^2$

C) $4.45\times10^5 N/m^2$ **D)** $5.45\times10^5 N/m^2$

2. Water at 100^0C is boiled in a rectangular steel tank of thickness 2 cm by a constant temperature furnace. Due to vaporization, water level falls at a steady rate of 1 cm in 9 minutes. If heat is supplied by furnace only from the bottom of tank then the temperature of furnace is (Given K for steel = 0.2cgs units, L_{vapour} = 540 Cal/gm).

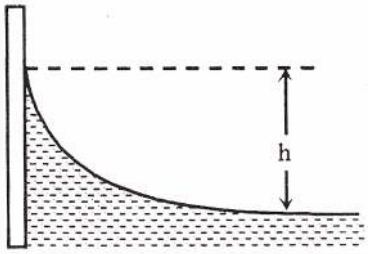
A) $150^0 C$ **B)** $120^0 C$ **C)** $110^0 C$ **D)** $200^0 C$

3. An electric bulb rated 100W- 300V is connected with an ac supply of 500 V and $\frac{150}{\pi}$ Hz. The required inductance to be connected in series to the bulb so that the bulb works at the rated power is

A) 2H **B)** H **C)** 4H **D)** 8H

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4. Water in a clean aquarium forms a meniscus, as illustrated in the figure. The difference in height (h) between the centre and the edge of the meniscus in mm is .... (nearly). The surface tension of water is 0.073 N/m.



A) 0.38 mm **B)** 3.8 mm **C)** 38 mm **D)** 0.038 mm

SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:*** **+2** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Question Stem for Question Nos. 5 and 6

Question Stem

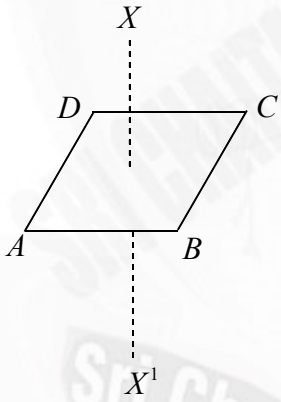


Figure shows a uniformly charged thin square sheet (ABCD) of side length 'a'. The net charge of sheet is Q. The axis XX^1 is perpendicular to the plane of sheet and passes through its centre.

5. The potential at corner A is V_0 in the above case. If side length is doubled keeping charge density same then potential at corner is nV_0. Find the value of 'n'.

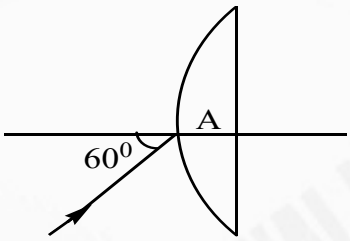
6. The electric field at a point on the axis XX^1 which is very close to centre and above the plane is $\frac{Q}{n\in_0 a^2}$. Find the value of 'n'.

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Question Stem for Question Nos. 7 and 8

Question Stem

Figure shows a plano convex lens of refractive index $\sqrt{3}$ placed in air. The maximum thickness of the lens is $3mm$ and its aperture diameter is $8mm$. Point A lies on the curved surface on the principal axis. A light ray is incident at the point A as shown in the figure making 60^0 with the normal. $\left(\sqrt{3}=1.732\right)$



7. The lateral displacement of the light ray in passing through the lens is (in mm)

8. The focal length of the lens if treated as a thin lens is (in mm)

Question Stem for Question Nos. 9 and 10

Question Stem

A monoenergetic beam of α - particles, each of them having kinetic energy E is incident on a sample of singly ionized Helium gas in ground state. The Helium sample may start emitting radiation since the kinetic energy of incident α - particles is sufficiently high. A part of this radiation (if any) is allowed to pass through atomic Hydrogen sample in ground state. A detector placed near the Hydrogen records both radiation and electrons. Assume all He^+ ions to be initially at rest. (ionization energy of H is 13.6 eV)

9. If maximum kinetic energy of electrons intercepted by the detector is 27.2 eV, the minimum kinetic energy of α - particle must be (in eV)

10. If maximum kinetic energy of electrons intercepted by the detector is 27.2 eV, the K.E. of α - particle must be lower than _____ eV

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

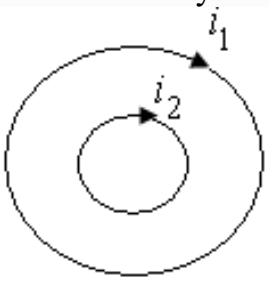
Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;

Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and Choosing any other option(s) will get -2 marks.

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11. Two concentric circular coils of radius R and r (r<<R) carry currents of i_1 and i_2 respectively. If the smaller coil of known mass is rotated slightly about one of its diameters, it starts oscillating. Then which of the following statements is/are incorrect. (assume that the currents in the coils are steady and larger coil is fixed)



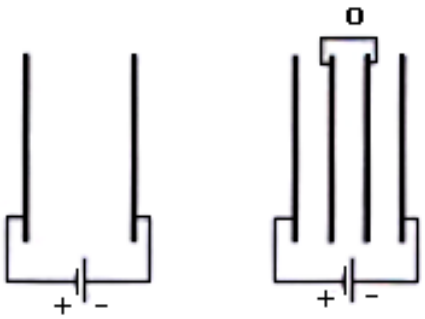
A) The oscillations are approximately simple harmonic in nature

B) The frequency of oscillations is proportional to product $i_1 i_2$

C) The frequency of oscillations is proportional to square root of R

D) The frequency of oscillations is proportional to radius r

12. A parallel plate capacitor is connected to a battery which builds up an electric field of $600\frac{V}{m}$ between the plates. Now two neutral plates are connected by a conducting wire and are placed into the capacitor as shown. The four plates are at equal distance from each other. Then



A) By introducing two plates, capacitance of system decreases and the battery absorbs energy

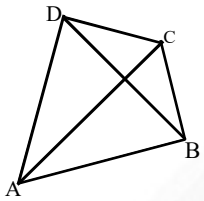
B) By introducing two plates capacitance increases and the battery supplies energy.

C) The electric field between pairs of plates from left to right is 900 V/m, zero, and 900 V/m respectively.

D) The energy of the system of capacitors increases.

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13. Six identical wires of resistance R each are joined as shown in the figure. Electrical contacts are only at A, B, C & D.



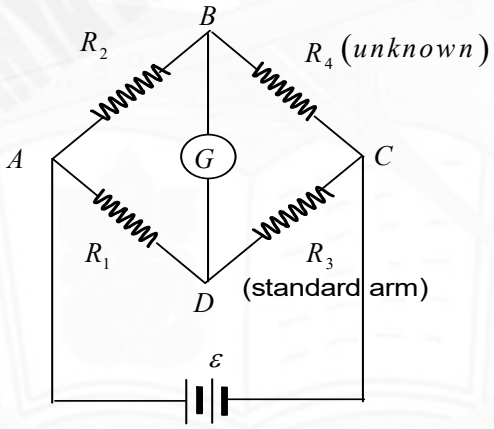
A) The equivalent resistance between any two corners will not depend on the choice of corners.

B) The equivalent resistance between A and B is $\frac{R}{2}$.

C) The equivalent resistance between D and C is zero.

D) If B and C are used as entry and exit points of current then there would be no current between A and D.

14. The measurement of an unknown resistance R_4 is to be carried out using Wheatstone bridge. Two students perform an experiment in two ways. The first students takes $R_2 = 10\Omega$ and $R_1 = 5\Omega$. The other student takes $R_2 = 1000\Omega$ and $R_1 = 500\Omega$. In the standard arm, both take $R_3 = 5\Omega$. Both find $R_4 = (R_2 / R_1) R_3 = 10\Omega$ within errors.



A) The errors of measurements of the two students are the same

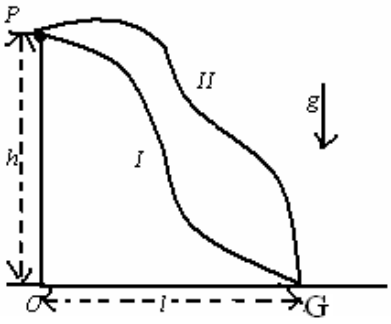
B) Errors of measurement do depend on the accuracy with which R_2 and R_1 can be measured

C) If the student uses large values of R_2 and R_1, the currents through the arms will be feeble. This will make accurate determination of null point more difficult

D) Wheatstone bridge is a very accurate instrument and has no errors of measurement

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15. In the figure shown P is the peak of a mountain. From the ground point G; a block of mass m and 2m are lifted slowly to the peak via path I and II respectively. Both paths have same co-efficient of friction. If the work done be respectively w_I and w_{II}; Choose the correct option/s



A) $2w_I = w_{II}$

B) $w_I = w_{II}$

C) loss of energy due to friction is same for both the paths

D) $w_I^1 = w_{II}^1$ (Where w_I^1 and w_{II}^1 are works done per unit mass)

16. When a floating object like a boat moves, it also has to move the water that it displaces. Due to this the 'effective mass' of the boat is increased by approximately one third of the mass of the boat. A boat of total mass M is moving at speed u and is accelerating at a rate a in a large lake. Pick the most appropriate option without neglecting viscosity of water.

A) Power delivered to boat is $\frac{2}{3}Mau$

B) Power delivered to water is $\frac{M}{3}au$

C) Total power delivered by motor of boat is greater than $\frac{4}{3}Mau$

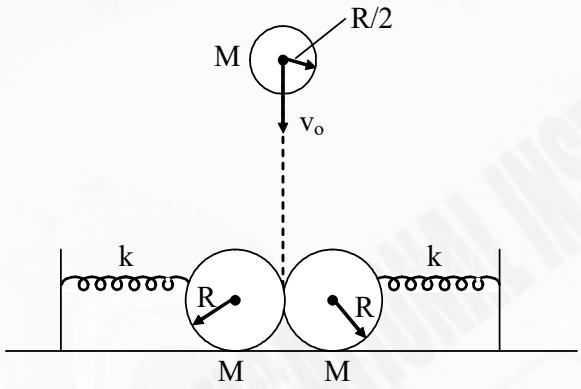
D) The net force acting on the boat is $M\sqrt{g^2+a^2}$

SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** : **+4** If ONLY the correct integer is entered;
- ***Zero Marks*** : **0** In all other cases.

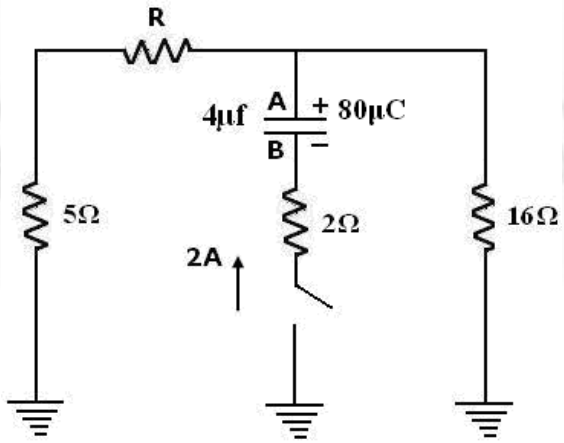
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17. Two identical balls each of mass M (=1.25 kg) and radius r are kept in contact and are connected to two identical unstretched springs each of force constant k(=100 N/m). A third ball also of mass M but radius R/2 strikes with the two balls symmetrically with speed v_0(=1.6m/s) and comes to rest just after comes to rest just after the impact. Find the maximum compression in each spring in cm. (Friction is absent everywhere. Ignore rotation of balls)



18. A point sound source is situated in a medium of bulk modulus $1.6\times10^5 N/m^2$. An observer standing at a distance 10 m form the source writes down the equation for the wave as $y = A\sin(15\pi x - 6000\pi t)$. Here y and x are in meters and t is in second. The maximum pressure amplitude received by the observer's ear is 24π Pa. Then the density of the medium is given by n kg/m^3. Find n.

19. An $80\mu C$ charge is given to the $4\mu F$ capacitor in the circuit shown in figure so that the upper plate is positively charged. An unknown resistance R is connected in the left limb. As soon as the switch 'S' in the central limb is closed, a current of 2A flows through the 2Ω resistor in the central limb. The capacitative time constant $(in\ \mu s)$ of the circuit is



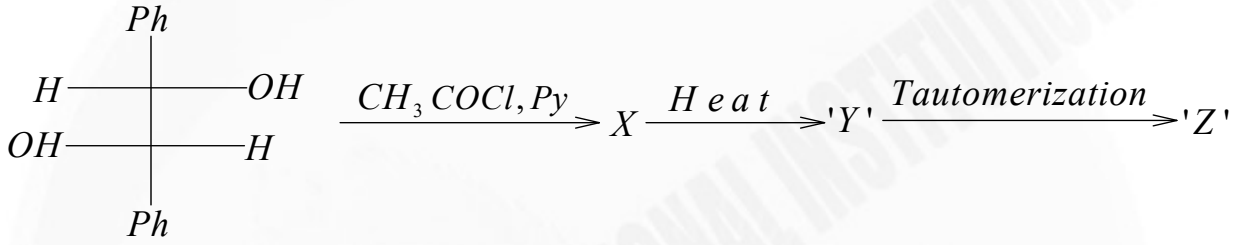
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CHEMISTRY **Max. Marks: 60**

SECTION 1

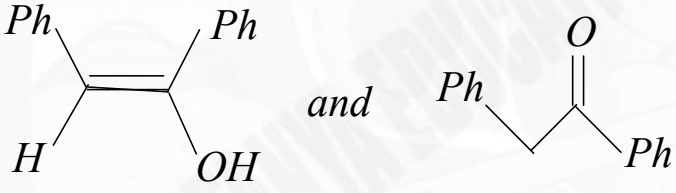
- This section contains **Four (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
- *Full Marks* : +3 If ONLY the correct option is chosen;
- *Zero Marks* : 0 If the none of the options is chosen (i.e. the question is unanswered);
- *Negative Marks* : -1 In all other cases.

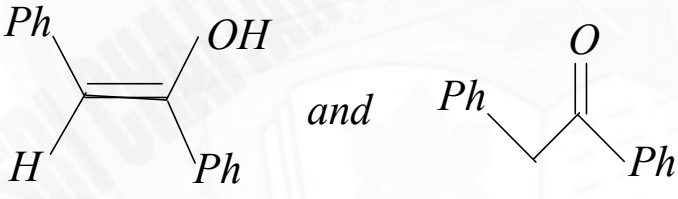
20.

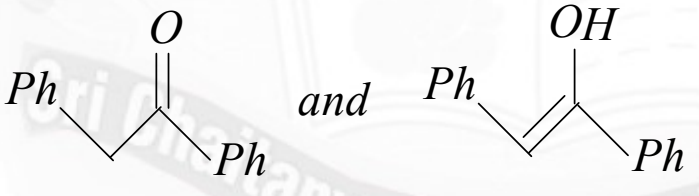


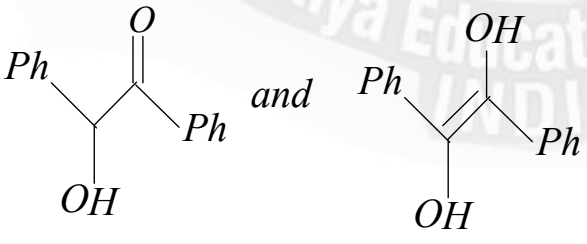
(Py = Pyridine)

If one equivalent of acetyl chloride is used, then main product, Y and Z are respectively

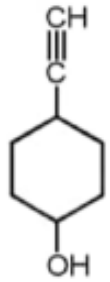
A) 

B) 

C) 

D) 

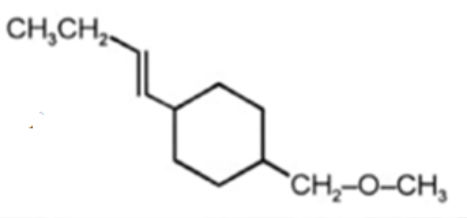
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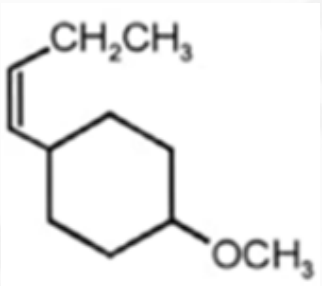


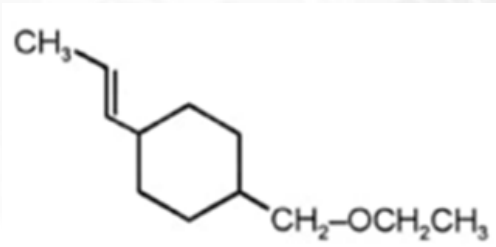
1. $NaNH_2$(2eq.)
2. CH_3CH_2–I
3. CH_3–I
4. H_2/Pd-$BaSO_4$

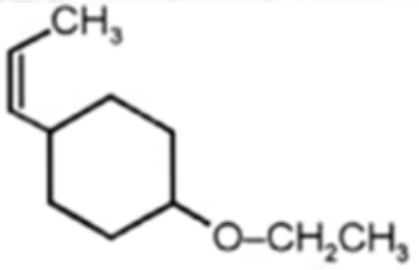
→ Product (X) is :

21.

A) 

B) 

C) 

D) 

22. Calculate the enthalpy of the following homogeneous gaseous reaction

$CH_3COCH_3 + 2O_2 \rightarrow CH_3COOH + CO_2 + H_2O$

from the following data.

Bond energies (kJ/mol): $C-H = 414$; $C-C = 348$; $C=O = 580$; $C-O = 354$;

$O=O = 610$; $O-H = 462$;

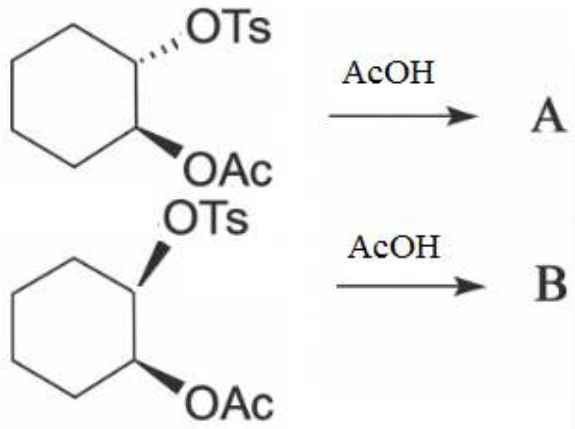
Magnitude of resonance energies (kJ/mol): $CH_3COOH = 118$; $CO_2 = 140$.

A) 348 kJ **B)** 168 kJ **C)** -168 kJ **D)** -348 kJ

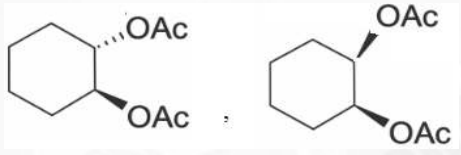
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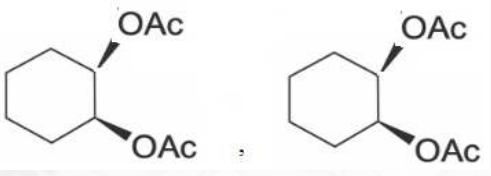

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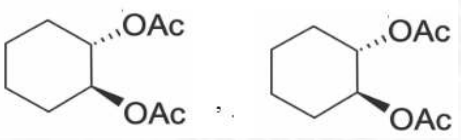
23.

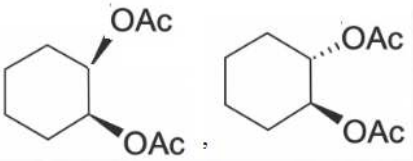


A and B are

A) 

B) 

C) 

D) 

SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +2** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks*** **: 0** In all other cases.

Question Stem for Question Nos. 24 and 25

Question Stem

At constant temperature and volume, X decomposes as $2X(g) \rightarrow 3Y(g) + 2Z(g)$. P_X is the partial pressure of X.

Observation No.	Time (min)	$P_X\,(mm\ Hg)$
1	0	800
2	100	400
3	200	200

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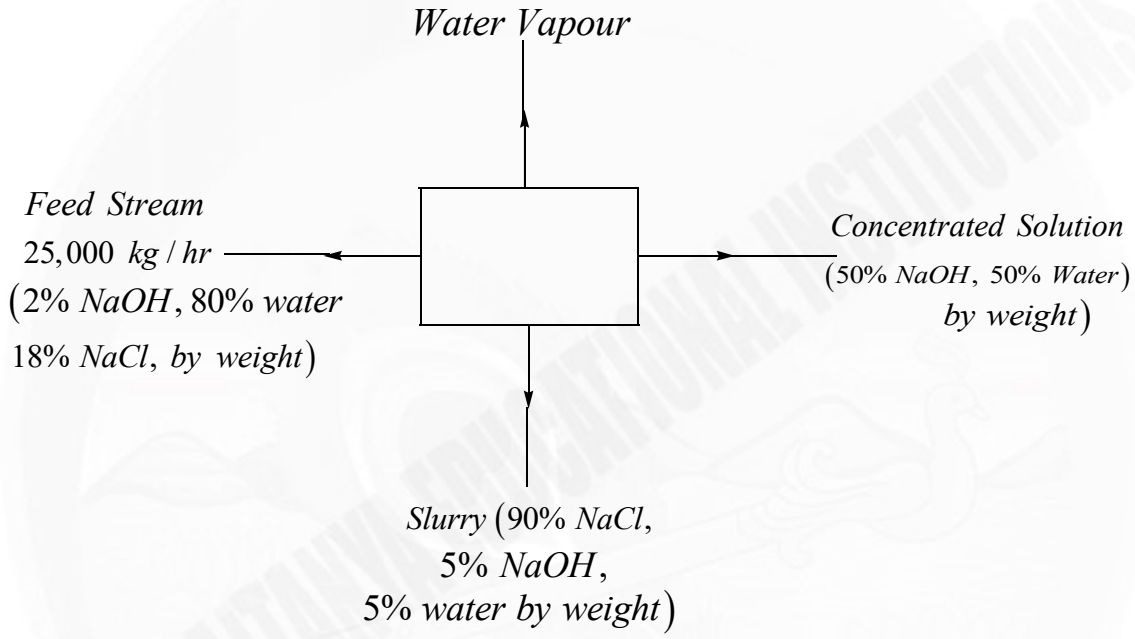
24. Time required for 87.5% completion of the reaction is_____min

25. The total pressure, when pressure of 'X' is 700 mm of Hg, is _____ mm of Hg,

Question Stem for Question Nos. 26 and 27

Question Stem

Figure shown a scheme for concentrating a dilute solution of $NaOH$.



26. The amount of water evaporated per hour in Kg is $Q \times 10$, The value of Q _____

27. The amount of concentrated solution obtained per hour in Kg is

Question Stem for Question Nos. 28 and 29

Question Stem

The wave function for an atomic orbital of single electron atom or ion is

$$\Psi(r,\theta,\phi) = \frac{2}{3}\left(\frac{Z}{3a_0}\right)^{3/2}(1-\sigma)\left(12-8\sigma+\sigma^2\right)\cdot\sigma\cdot e^{-\sigma/2}\cdot\cos\theta$$

Where $\sigma = \dfrac{2Zr}{na_0}$ and $a_0 = 0.59\,A^\circ$. All other parameters have their usual meaning.

28. The number of radial and angular nodes for the orbital is, respectively, x and y the value of $x + y$ is

29. The $n + \ell$ value of the atomic orbital should be

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SECTION 3

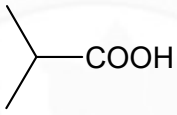
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

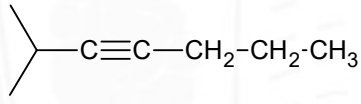
Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

30. An organic compound **P** of molecular formula C_5H_8, when treated with Na in liquid NH_3 followed by reaction with n–propyl iodide yields **Q** (C_8H_{14}). **P** gives a ketone **R** $C_5H_{10}O$, when treated with dilute H_2SO_4 and $HgSO_4$. **Q** On oxidation with alkaline $KMnO_4$ gives two isomeric acids **S** and **T** of formula $C_4H_8O_2$. The correct statement(s) among the following regarding this reaction sequence is/are

A) Compound **R** is C_3H_7–CO–CH_3 and S may be C_3H_7COOH

B) Compound **P** is $C_2H_5C \equiv CCH_3$

C) Compound **T** may be 

D) Compound **Q** may be 

31. Which of the following statements is/are correct

A) $NH_3 > SbH_3 > AsH_3 > PH_3$ (melting point order)

B) According to Valence bond theory in NH_3 and H_2O molecules, the HNH and HOH angles should be $90°$ [considering pure atomic orbitals].

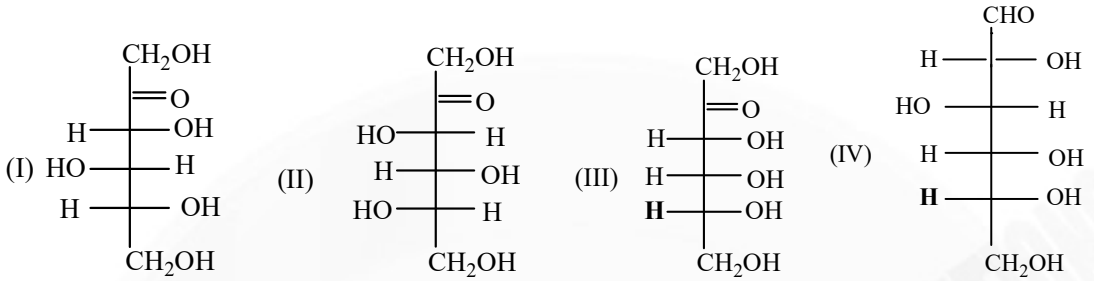
C) In ethene the $H-C-H$ bond angle is less than $120°$ while the $H-C-C$ angle is more than $120°$.

D) The $C-O-C$ bond angle in methoxy methane is less than $109.5°$.

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32. Which relationship (s) is/are true :



A) $NaBH_4$ reduce IV to one polyalcohol where as I,II & III two polyalcohols each

B) II and III are diastereomers

C) I and III are epimers

D) I,II,III and IV forms different osazones

33. Which of the following statements is/are correct?

A) The mineral $KCl.MgCl_2.6H_2O$ is known as carnallite.

B) The iron obtained from Blast furnance is known as pig iron which contains about 4% carbon and many impurities (S, P, Si, Mn) in smaller amount.

C) Froth floatation method of concentrating the ore is used for removing gangue from sulphide ores.

D) Silver and gold are extracted from their native ore by the hydro metallurgical process.

34. The correct information regarding a solid element (atomic radius = r) having ABCABC type packing (FCC unit cell edge length = a) is:

A) the distance between nearest octahedral and tetrahedral void is $\frac{a\sqrt{3}}{4}$

B) the distance between two nearest octahedral void is $\frac{a}{\sqrt{2}}$

C) the distance between two nearest tetrahedral void is $\frac{\sqrt{3a}}{4}$

D) the distance between successive A and B layers is $2r\sqrt{\frac{2}{3}}$

35. Which of the following set of molecules have same EAN value as $Ni(CO)_4$

A) $V(CO)_5, Cr(CO)_5, Cu(CO)_3$

B) $Fe(CO)_5, Mn(CO)_5, Cu(CO)_3$

C) $[Cr(CO)_4]^{4-}, [Co(CO)_3]^{3-}, Fe(CO)_5$

D) $[Mn_2(CO)_{10}], [Co_2(CO)_8], [Fe(CN)_6]^{4-}$

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SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

36. A vessel of 2.5 L was filled with 0.01 mole of Sb_2S_3 and 0.01 mole of H_2 to attain equilibrium at 440^0 C as $Sb_2S_3(s)+3H_2(g)\rightleftarrows 2Sb(s)+3H_2S(g)$

After equilibrium, the H_2S formed was analysed by dissolving it in water and treating with excess of Pb^{2+} to give 1.19g of PbS as precipitate. The value of K_c of the reaction at 440^0 C is (Awt of Pb = 206)

37. 100 mL of 0.05 M $CuSO_4(aq)$ solution was electrolyzed using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess KI and formed I_2 titrated with 0.04 M $Na_2S_2O_3$. If the required volume (in mL) of $Na_2S_2O_3$ is P, the value of $\frac{P}{10}$ is _____

[Round of the value of $\frac{P}{10}$ to nearest integer]

38. A volume of 224 ml of $CO_2(g)$ at 1 atm and 0^0C was passed in 1 L of $NaOH$ solution of unknown molarity. The resulting solution when titrated with 1.0 M-HCl solution required 30 ml for the phenolphthalein end point. The mass of NaOH in milligrams dissolved in 1L of the NaOH solution used is a, the value of $\frac{a}{100}$ is

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MATHEMATICS **Max. Marks: 60**

SECTION 1

- This section contains **Four (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
- *Full Marks* : +3 If ONLY the correct option is chosen;
- *Zero Marks* : 0 If the none of the options is chosen (i.e. the question is unanswered);
- *Negative Marks* : -1 In all other cases.

39. Considering only the principal values of the inverse trigonometric functions, the value of $\tan\left(\sin^{-1}\left(\frac{3}{5}\right)-2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$ is

A) $\frac{-7}{24}$ **B)** $\frac{7}{24}$ **C)** $\frac{-5}{24}$ **D)** $\frac{5}{24}$

40. If $A=\left[a_{ij}\right]_{2\times 2}$, $B=\left[b_{ij}\right]_{2\times 2}$ *such that* $a_{ij}=f\left(x^{2i-j-2}\right)$ and $b_{ij}=f\left(x^{2i-j-1}\right)$ and

$$A+B^T=\begin{bmatrix}-x^2 f\left(x+f\left(1\right)\right) & \left(1-\frac{1}{x^4}\right)f\left(\frac{1}{x^2}\right)\\ f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right) & -\frac{1}{x^2}f\left(\frac{1+f\left(1\right)}{x}\right)\end{bmatrix}$$, Then $\int_x^{\frac{1}{x}} f\left(t\right)dt$ is always

equal to

A) $f\left(1\right)$ **B)** $f\left(1\right)-1$ **C)** $f\left(1\right)+1$ **D)** -7

41. Let $d\left(P,OA\right)\le \min\left\{d\left(P,\,AB\right),d\left(P,\,BC\right),d\left(P,\,OC\right)\right\}$ where d denotes the distance from the point to the corresponding line and S be the region consisting of all those points P inside the rectangle OABC with vertices $O\left(0,\,0\right)$, $A\left(3,\,0\right)$, $B\left(3,\,2\right)$ and $C\left(0,\,2\right)$; which satisfy the above relation. Then area of the region 'S' is

A) 2 sq units **B)** 3 sq units **C)** 4 sq units **D)** 5 sq units

42. The system of equations

$\log_{10}\left(2000xy\right)-\log_{10}x.\log_{10}y=4$, $\log_{10}\left(2yz\right)-\log_{10}y.\log_{10}z=1$,

$\log_{10}\left(zx\right)-\log_{10}z.\log_{10}x=0$ has two solutions $\left(x_1,\,y_1,\,z_1\right)$ and $\left(x_2,\,y_2,\,z_2\right)$. Then

y_1+y_2 is equal to

A)10 **B)** 15 **C)** 20 **D)** 25

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SECTION 2

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

Question Stem for Question Nos. 43 and 44

Question Stem

For $a,b\in\mathbb{R}-\{0\}$, Let $f(x)=ax^2+bx+a$ satisfies $f\left(x+\frac{7}{4}\right)=f\left(\frac{7}{4}-x\right)\forall x\in\mathbb{R}$. Also the equation $f(x)=7x+a$ has only one real solution then

43. $\frac{a+b}{2}=$ _____

44. $\left|\frac{\text{min}\,imum\ value\ of\ f(x)\ in\left[0,\frac{3}{2}\right]}{5}\right|=$ _____

Question Stem for Question Nos. 45 and 46

Question Stem

Let a straight line L_1 whose slope is negative, passing through a point $A(0,8)$, another line perpendicular to L_1, meeting it at 'P' and passing through $B(0,-2)$.

45. If abscissa of point P is maximum, then area of ΔAPB is –

46. If incentre of ΔAPB is $\left(\frac{5\sqrt{3}-5}{2},\frac{1+5\sqrt{3}}{2}\right)$ then equation of line L_1 is in the form of $x+\sqrt{a}y=b\sqrt{a}$ then $a+b$ is equal to

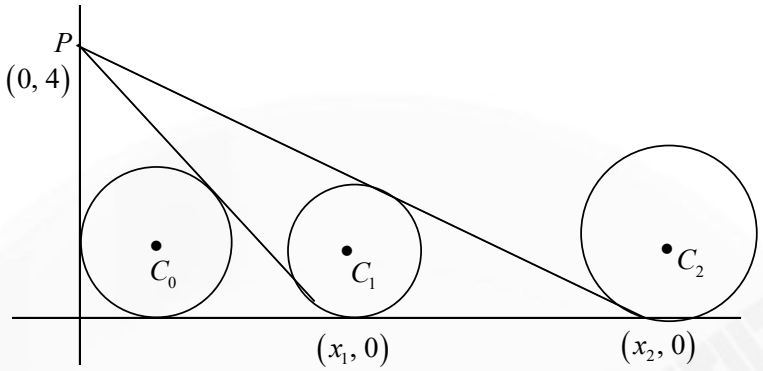
Question Stem for Question Nos. 47 and 48

Question Stem

If a circle C_0 with radius 1 unit touches both axes and as well as line (L_1) through $P(0,4)$, cut the x-axis at $(x_1,0)$. Again a circle C_1 of radius one unit is drawn touching x-axis, line L_1 and another line L_2 through point P. L_2 intersect x-axis at $(x_2,0)$ and this process is repeated 'n' times as shown.

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47. The centre of circle is C_2 (p, q) Then, the value of $[p+q]$ is equal to _____

Here [.] denote greatest integer function

48. $\lim_{n\to\infty} \frac{x_n}{2^n}$ is equal to

SECTION 3

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks*** **: +3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks*** **: +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks*** **: +1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks*** **: 0** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

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49. Which of the following is/are correct?

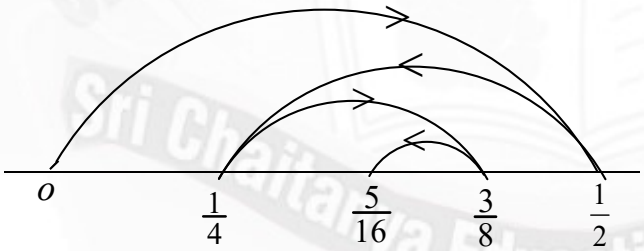
A) If the ellipse $x^2+\lambda^2y^2=\lambda^2a^2;\ \lambda^2>1$ is confocal with hyperbola $x^2-y^2=a^2$, then ratio of eccentricity of ellipse to eccentricity to hyperbola is $1:\sqrt{3}$

B) The asymptote of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ form with any tangent to the hyperbola a triangle whose are is $a^2\tan\lambda$ in magnitude, then its eccentricity is $\sec\lambda$

C) If the circle $x^2+y^2=a^2$ intersects the hyperbola $xy=c^2$ at four points $P(x_1,y_1);Q(x_2,y_2);R(x_3,y_3)$ *and* $S(x_4,y_4)$ *then* $\sum x_i+\sum y_i+\prod x_i+\prod y_i=4c^4$ where i = 1, 2, 3, 4

D) If the line $y=mx+\sqrt{a^2m^2-b^2}$ touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at the point $(a\sec\theta,b\tan\theta)$ the $\theta=\sin^{-1}\left(\frac{b}{am}\right)$

50. A flea to amuse itself jumps $\frac{1}{2}$ meter to right then $\frac{1}{4}$ meter to left then $\frac{1}{8}$ meter to right then $\frac{1}{16}$ meter to the left and so on as shown in the figure. If after n jumps the flea is at t_n then



A) $t_9=\left(\frac{171}{512}\right)$ **B)** $t_{10}=\frac{341}{1024}$

C) $\sum_{r=1}^{n}t_r=\frac{n}{3}+\frac{1}{9}\left(\frac{2^n+1}{2^n}\right)$, When n-odd **D)** $\sum_{r=1}^{n}t_r=\frac{n}{3}+\frac{1}{9}\left(\frac{2^n-1}{2^n}\right)$, When n-even

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51. Which of the following is/are correct?

A) A one to one mapping f is selected at random from set

$A=\{x_1,x_2,x_3,x_4,x_5\}$ to the set $B=\{y_1,y_2,y_3,y_4,y_5\}$

Then the probability that the selected mapping f is such that $f(x_i)\neq y_i$ for $i=1,2,3,4,5$

is $\frac{11}{30}$

B) Of all the mappings that can be defined from the set $A=\{1,2,3,4\}$ to the set $B=\{5,6,7,8,9\}$, a mapping a selected at random, then the probability that the selected mapping is strictly monotonic is $\frac{2}{125}$

C) Let the set A have 7 elements and the set B have 5 elements. If one function is selected from all possible defined functions from A to B then the probability that it is onto is $\frac{7!\times 2}{5^6\times 3}$

D)If a function 'f' from the set $A=\{0,1,2,\}$ into the set $B=\{0,1,2,3,4,5,6,7\}$ is selected then the probability of 'f' such that $f(i)\leq f(j)$

For $i<j$ and $i,\ j\in A$ is $\frac{15}{64}$

52. Let $S_1=\{(i,j,k):i,j,k\in\{1,2,....,10\}\}$,

$S_2=\{(i,j):1\leq i<j+2\leq 10,i,j\in\{1,2,....,10\}\}$,

$S_3=\{(i,j,k,l):1\leq i<j<k<l,i,j,k,l\in\{1,2,....,10\}\}$,

and $S_4=\{(i,j,k,l):i,j,k$ and l are distinct elements in $\{1,2,...,10\}\}$. If the total number of elements in the set S_r is n_r $r=1,2,3,4,$ then which of the following statements is (are) TRUE?

A) $n_1=1000$ **B)** $n_2=44$ **C)** $n_3=220$ **D)** $\frac{n_4}{12}=420$

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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1 300
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
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53. If $S=\log_e\left(1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{100}\right)$ and $T_n=\frac{1}{\log_e\left(\prod_{r=1}^{n}\frac{r^2}{1+r^2}\right)}$

Then which of the following statement(s) is/are true? (Note: $l\,n(x)=\log_e x$, $\log_e 2\cong 0.69,\ \log_e 3\cong 1.098, \log_e 4\cong 1.38,\ \log_e 5\cong 1.60$ & $\log_e 6\cong 1.79,\ldots..)$

A) $T_n>\frac{4n}{(n-1)^2-n^2\log_e 4}$ **B)** $S<\frac{100}{99}\log_e 50$

C) $T_n<\frac{2}{n(1-\log_e 2)-\cos^{-1}\left(\frac{2n}{1+n^2}\right)}$ **D)** $S+T_n>\frac{4n+100}{\frac{99}{100}+n\log_e 4}$

54. Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function such that

$$f(x)=\int_0^x e^{x-y}f'(y)dy-\left(x^2-x+1\right)e^x$$

for all $x\in\mathbb{R}$. Then, which of the following statement(s) is/are TRUE?

A) f has a local minimum at $x=-1/2$

B) $f(x)>1-x+\frac{3x^2}{2}-\frac{5x^3}{6}$ for all $x\in(0,\infty)$

C) $\int_0^1 f(x)dx=\int_0^1 f(1-x)dx$

D) For every $x>0$, there exists a $\beta\in(0,x)$ such that $f'(\beta)=2e^{\beta}$

SECTION 4

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

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55. The curve $y=(|x|-1)\,sgn(x-1)$ Divides $\frac{9x^2}{64}+\frac{4y^2}{25}=\frac{1}{\pi}$ into two parts areas A_1 and A_2 $(A_1<A_2)$, Then $A_1:A_2=p:q$ & G.C.D of $(p,q)=1$. Then $p+q$ is equal to

56. If the number of ordered Triplets (a, b, c) such that

L.C.M (a, b) = 1000

L.C.M (b, c) = 2000 and L.C.M of (c, a) = 2000 be M, then the sum of digits of M is

57. Let the function $f:\mathbb{R}\to\mathbb{R}$ defined by

$$f(x)=\frac{\sin x}{e^{\pi x}}.\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}+\frac{2}{e^{\pi x}}.\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)},$$ then the number of solutions of $f(x)=0$ in $\mathbb{R}$, is _____ ($\mathbb{R}$ denotes set of real numbers)

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APP. NO. 250310255592
CLASSROOM STUDENT FROM GRADE IX - XII



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APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100	BELOW 500	BELOW 10	BELOW 100	BELOW 1000
31	95	10	98	579
ALL INDIA OPEN CATEGORY RANKS	ALL INDIA OPEN CATEGORY RANKS	ALL INDIA CATEGORY RANKS COUNT	ALL INDIA CATEGORY RANKS COUNT	ALL INDIA CATEGORY RANKS COUNT

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

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ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
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H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

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